

From Distinction to Statistics: A Disciplined Companion Essay

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Abstract

This companion essay restates, in a wider register, the thesis of the paper “The Boson/Fermion Distinction: Spin-Statistics as Structural Invariant” (DOI 10.5281/zenodo.21938971). It separates three things that are often conflated in foundational writing: (1) what is established, (2) what is a pre-registered program, and (3) what is philosophical extrapolation. The distinction is kept explicit throughout.

1. What is established

The exchange phase of identical particles equals their topological spin:

$$R = e^{2\pi i s}.$$

[ESTABLISHED] This relation holds in relativistic quantum field theory, in topological field theory, and for fractional quantum Hall quasiparticles, where fractional spin is measurable. In three or more spatial dimensions the braiding is involutive, so $R = \pm 1$ and the spin parameter is quantized to integers and half-integers — the boson/fermion dichotomy is a shadow of one relation. In two spatial dimensions the braid group permits any phase, giving anyons.

The invariant formulation itself earns no predictive credit: it restates results that are already established in three independent research traditions. It is therefore labeled [RETRODICTION — not evidence].

2. The honest boundary

The paper states, and this essay repeats, the boundary of the derivation program: the mark calculus can force *statistics* (the two eigenvalues of exchange) from the act of distinction, but it cannot, from the mark alone, derive the *spin–statistics connection* — the identification of which eigenvalue maps to which spin. That identification

requires the additional postulate that the twist equals the 2π rotation of a Lorentz representation, together with microcausality and positive energy. Any account that omits this boundary overstates the program.

3. The derivation program (pre-registered; structural checks executed)

Three tasks, with falsifiability conditions:

- T1: two modal exponentials, the symmetric and exterior algebras, as the two grading components of the exchange operator on the mark.
- T2: the braiding of two self-dual marks, whose eigenvalues are the two statistics.
- T3: the dimensional table relating allowed values of the spin parameter to the braided structure.

The program is [NOT YET EVIDENCE]: it is pre-registered, and the toy-model demonstration (a pre-registered syntactic check) is a syntactic confirmation, not a physical derivation. The T1/T2 structural check (two modal exponentials as DiLL modalities; braiding of two self-dual marks; ribbon identity; abelian-pair postulate) has since been executed at the notebook level — a structural verification, not a physical derivation. Falsifiability: F1 — an observed stable local relativistic 3+1D excitation with exchange phase not equal to $e^{2\pi i s}$ would disconfirm the invariant; F2 — an impossibility proof showing the two one-dimensional characters of the symmetric group cannot be recovered from the primitive alone would disconfirm the program.

4. What is extrapolation (not established)

The following claims, which appear in the surrounding deep-inquiry literature, are philosophical extrapolations and are labeled as such here. They are not part of the published paper and carry no evidential weight:

- [EXTRAPOLATION] “The universe is made of distinctions, not matter.”
- [EXTRAPOLATION] Mass, charge, and spin are braid invariants of a boundary’s worldline.
- [EXTRAPOLATION] Spacetime dimension is emergent from the weaving of distinctions.
- [EXTRAPOLATION] “Bit from distinction” extends Wheeler’s “It from Bit.”

These are coherent and worth pursuing, but they are metaphysics until they yield falsifiable predictions. Any published form must carry this label.

5. External constraints (not to be omitted)

The invariant reading must be read against live alternatives and constraints:

- A dynamical origin for the exchange phase via gravity has been proposed (Marletto and Vedral, 2021, arXiv:2112.03392) — a genuine alternative mechanism.

- The standard derivations are not uncontested: Jabs (arXiv:quant-ph/0311078) and Lev (arXiv:hep-th/0212178) challenge aspects of the standard proofs.
- Unphysical sectors already violate the naive rule: Faddeev–Popov ghosts are scalar yet anticommuting, so the claim is scoped to physical states.

6. The silo it rectifies

A quantified consilience finding: the Laws of Form tradition (Spencer-Brown, 1969) and the spin-statistics theorem (Pauli, 1940) developed with no contact for over fifty years, and no published derivation of exchange statistics from the mark calculus exists as of this writing. The paper’s derivation target is, to the author’s knowledge, the first explicit statement of that connection.

7. Open frontier question

The mark calculus treats boundary-drawing as a free primitive. Thermodynamics suggests otherwise: creating a distinction costs free energy (Landauer’s principle). If the primitive act carries a cost, then energy and entropy may be more primitive than the distinction itself, and the second law may precede the first cut. This is the strongest open question raised by the program, and it is registered as an open frontier question in the project’s continuity registry, with a pre-registered falsification condition.

Declarations

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